**This assignment will not be graded and is only for practice.**

1) Let $L_1$ and $L_2$ be affine subspaces of $\mathrm{R}^3$, where $L_1 = U$ and $L_2 = (2, 0, 1) + W$, for some vector subspaces $U$ and $W$ of $\mathrm{R}^3$. Let a basis for $U$ be given by $\{(2, 0, 1), (1, 1, 0), (0, 1, 0)\}$ and a basis for $W$ be given by $\{(1, 0, 1), (0, 1, 1)\}$. Suppose there is a linear transformation $T : U \to W$ such that $(0, 1, 0) \in ker(T)$, $T(2, 0, 1) = (0, 1, 1)$ and $T(1, 1, 0) = (1, 0, 1)$. An affine mapping $f : L_1 \to L_2$ is obtained by defining $f(u) = (2, 0, 1) + T(u)$, for all $u \in U$. Which of the following options are true? ***1 point***

- [ ] $L_1 = \mathrm{R}^3$
- [ ] $L_2 = \{(x, y, z) \mid x + y - z = 1\}$
- [ ] $L_2 = \mathrm{R}^3$
- [ ] $f(x, y, z) = (x - 2z + 2, z, x - z + 1)$
- [ ] $f(x, y, z) = (x - 2z, z, x - z)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$L_1 = \mathrm{R}^3$
$L_2 = \{(x, y, z) \mid x + y - z = 1\}$
$f(x, y, z) = (x - 2z + 2, z, x - z + 1)$

**Solution:**

**Recall the definition of Affine map:**

Let $L_1$ and $L_2$ be affine subspaces of $V_1$ and $V_2$, respectively, and let $U$ and $W$ be the unique subspaces that correspond to $L_1$ and $L_2$, respectively.

- A map $f : L_1 \to L_2$ is an affine map if there exists a linear transformation $T : U \to W$ such that

$$f(v_1 + u) = v_2 + T(u),$$

where $v_1$ and $v_2$ are elements of $V_1$ and $V_2$, respectively such that $L_1 = v_1 + U$ and $L_2 = v_2 + W$.